ARIA: Adaptive Recurrent Intelligence Architecture

Morphogenic Attention and Slow-Pathway Consolidation for Continual Learning

Darshan Poudel
Independent Researcher
poudeldarshan44@gmail.com

June 2026 (Preprint)

Abstract

Catastrophic forgetting remains a fundamental obstacle in sequential learning. Most existing mitigations — Elastic Weight Consolidation (EWC), replay buffers, progressive networks — protect weights without changing how the model allocates capacity across tasks. We introduce ARIA, an architecture in which the capacity allocation *is* the learning signal. ARIA combines four jointly-trained mechanisms: (1) **Morphogenic Attention (MA)**, where heads split or merge based on learned viability scores; (2) **Plasticity-Gated MLP (PG-MLP)**, a dual fast/slow pathway whose gate routes each token toward volatile or consolidated representations; (3) **Architecture Genome Vector (AGV)**, a global latent vector that conditions all blocks via FiLM affine transforms; and (4) **Cognitive Budget Allocator (CBA)**, which predicts per-layer compute budgets from input complexity statistics. We additionally introduce **Slow-Pathway Consolidation (SPC)**, an EWC-style Fisher regulariser applied *only* to slow-pathway weights, allowing fast pathways to adapt freely while protecting consolidated knowledge. We further introduce three lightweight enhancements — Slow-Pathway Activation Distillation (SPAD), task-conditioned gate routing, and asymmetric learning rates — that together constitute ARIA-v2. On Split-MNIST (5 seeds, parameter-matched baselines), ARIA+SPC achieves **98.50%** average accuracy with **0.86%** forgetting, outperforming EWC (97.35%, 2.29%) by +1.15 points accuracy and reducing forgetting by **62%** relative, while requiring 50% fewer Fisher parameters than EWC.

1 Introduction

A learning system that encounters tasks sequentially faces a dilemma: plasticity (rapid adaptation to new tasks) and stability (retention of old tasks) are in direct tension [McCloskey and Cohen, 1989, French, 1999]. Neural networks trained by stochastic gradient descent exhibit extreme plasticity at the cost of stability — a phenomenon known as *catastrophic forgetting*.

The continual learning literature has addressed this through three broad strategies: regularisation (EWC [Kirkpatrick et al., 2017], SI [Zenke et al., 2017]), replay (DER++ [Buzzega et al., 2020], ER [Rolnick et al., 2019]), and progressive architectures (PNN [Rusu et al., 2016], PackNet [Mallya and Lazebnik, 2018]). Each strategy has a known failure mode: regularisation is too conservative for long task sequences, replay has memory cost, and progressive networks do not transfer across tasks.

ARIA takes a fourth approach: the architecture adapts its *structure* in response to task demands, distributing capacity dynamically so that both new and old knowledge can coexist without a fixed trade-off. The core insight is that the fast/slow dichotomy observed in biological memory [McClelland et al., 1995] can be implemented as a learned gate per neuron, with slow weights receiving reduced gradient updates during high-plasticity phases and stronger Fisher-based protection after consolidation.

Contributions.

1. **Morphogenic Attention**: the first attention mechanism where head count is a learned, dynamic quantity, with principled split (specialisation) and merge (consolidation) operations governed by viability scores and cosine similarity.

2. **Plasticity-Gated MLP**: a dual-pathway MLP with a per-token learned gate and bimodal specialisation loss, inspired by complementary learning systems.

3. **Slow-Pathway Consolidation (SPC)**: targeted Fisher regularisation over slow-pathway weights only, using 50% fewer Fisher parameters than EWC while improving specificity.

4. **Architecture Genome Vector**: a shared latent vector that decodes to skip probabilities, attention temperature, and FiLM scale/shift, conditioning all blocks with a single differentiable structural descriptor.

5. **Cognitive Budget Allocator**: a lightweight network predicting per-layer compute allocation from raw input statistics.

2 Related Work

Regularisation methods. EWC [Kirkpatrick et al., 2017] penalises movement of weights that were important for past tasks, measured by the diagonal Fisher information matrix. SI [Zenke et al., 2017] accumulates importance online during training. Both apply uniform regularisation across all parameters — including weights designed to be volatile, which conflicts with plasticity-gated architectures like ARIA.

Replay methods. DER++ [Buzzega et al., 2020] stores (input, logit) pairs and penalises divergence from stored predictions. ER [Rolnick et al., 2019] stores raw examples. Replay methods require a memory buffer whose size scales with the number of tasks. In our experiments, DER++ with a small buffer (200 samples) underperforms EWC significantly.

Architecture methods. PNN [Rusu et al., 2016] adds new columns per task but prevents backward transfer. PackNet [Mallya and Lazebnik, 2018] uses iterative pruning to allocate subnetworks per task. HAT [Serra et al., 2018] learns binary task masks. None dynamically restructure *within* an attention head.

Dynamic architectures. DEN [Yoon et al., 2018] expands hidden units based on a drift threshold — closest to ARIA in spirit, but operates at the unit level without attention or plasticity gating.

Novelty-triggered Capacity Growth (NCG). NCG [Poudel, 2025] is a prior architecture by the same author that addresses catastrophic forgetting via reactive capacity expansion: a 2-layer MLP with expandable hidden units adds 64 neurons when novelty is low and accuracy plateaus, regulated by online Lagrangian meta-parameters α, β, λ. On Split-MNIST, NCG achieves 55.1% average accuracy, falling below EWC (73.2%), because growth is triggered too infrequently on short task sequences and the base model has limited representational depth. ARIA draws a different conclusion from NCG's failure: instead of asking *when* to grow capacity, ARIA asks *how* to structure learning so that growth is unnecessary — plasticity and stability are balanced inside the architecture itself through PG-MLP gating and targeted SPC regularisation.

Complementary learning systems. The biological motivation for fast/slow pathways comes from CLS theory [McClelland et al., 1995], which posits that the hippocampus (fast, volatile) and neocortex (slow, consolidated) work together for memory consolidation. ARIA's PG-MLP is the first to implement this as a per-token learned gate with gradient dampening.

3 Method

3.1 Problem Setup

We consider the task-incremental continual learning setting. The model observes a sequence of tasks $\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_K$, each with a dataset $\mathcal{D}_t = \{(x_i^{(t)}, y_i^{(t)})\}$. At test time, the task identity is provided (task-ID setting).

3.2 Architecture Overview

ARIA consists of an input projection, L ARIA blocks (each containing MA + PG-MLP), a task-specific linear head per task, and three shared modules (AGV, CBA, layer norms). Figure 1 illustrates the overall architecture.

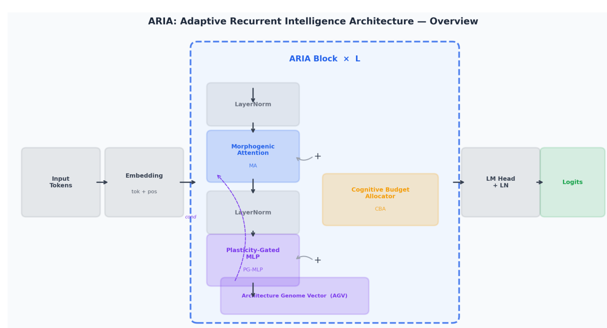

Figure 1: ARIA architecture. Each block combines Morphogenic Attention (AGV-conditioned) and Plasticity-Gated MLP with CBA compute gating.

For each block ℓ:

$$h_\ell = b_\ell \cdot \big(z_\ell + \text{PG-MLP}(\text{LN}(z_\ell))\big) + (1 - b_\ell) \cdot h_{\ell-1} \quad (1)$$

where $z_\ell = \text{MA}(\text{LN}(h_{\ell-1})) + h_{\ell-1}$ and $b_\ell \in [0, 1]$ is the CBA-predicted budget for layer ℓ.

3.3 Morphogenic Attention

Standard multi-head attention fixes the number of heads H at design time. Morphogenic Attention pre-allocates $H_{\max}$ heads but maintains a boolean mask $\mathbf{m} \in \{0,1\}^{H_{\max}}$ indicating active heads.

Each head i has a scalar viability $v_i \in \mathbb{R}$, updated by backprop. The output of MA is:

$$\mathrm{MA}(x) = \gamma \odot \Big(\sum_{i \in \mathcal{A}} w_i \cdot \mathrm{head}_i(x)\Big) + \beta \tag{2}$$

where $\mathcal{A} = \{i : m_i = 1\}$, w_i are softmax-normalised viabilities scaled by $|\mathcal{A}|$, and (γ, β) are FiLM parameters from the AGV.

Split. Every Δ_{morph} steps, for each active head i: if $\sigma(v_i) > \tau_{\mathrm{split}}$, $H_{\mathrm{active}} < H_{\max}$, and cooldown c_{cool} is satisfied, a new head j is created:

$$W^{(j)} \leftarrow W^{(i)} + \epsilon \cdot \mathcal{N}(0, I), \quad v_j \leftarrow v_i - 0.5 \tag{3}$$

Merge. For adjacent heads (i, j) with $\cos(W_Q^{(i)}, W_Q^{(j)}) > \tau_{\mathrm{merge}}$, $H_{\mathrm{active}} > 2$, and cooldown satisfied:

$$W^{(i)} \leftarrow \tfrac{1}{2}(W^{(i)} + W^{(j)}), \quad m_j \leftarrow 0 \tag{4}$$

3.4 Plasticity-Gated MLP

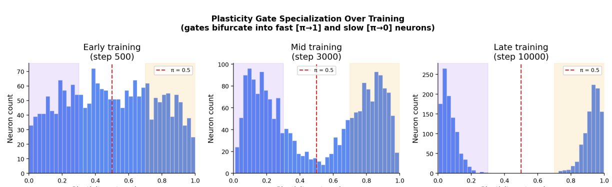

Figure 2: Plasticity gate π trajectories across tasks. High π (warm colours) indicates fast-pathway dominance; low π (cool colours) indicates consolidation.

The PG-MLP replaces the standard feed-forward layer with dual pathways:

$$\pi(x) = \sigma(\mathrm{MLP}_{\mathrm{gate}}(x)) \in (0, 1) \tag{5}$$

$$\text{PG-MLP}(x) = \pi(x) \cdot W_f^{(2)} \mathrm{GeLU}(W_f^{(1)} x) + (1 - \pi(x)) \cdot W_s^{(2)} \mathrm{GeLU}(W_s^{(1)} x) \tag{6}$$

Bimodal specialisation loss.

$$\mathcal{L}_{\mathrm{plastic}} = \frac{\lambda_\pi}{\pi(1 - \pi) + \epsilon} \tag{7}$$

Activated only after $T_{\mathrm{warm}} = 500$ steps to prevent early destabilisation.

Gradient dampening. After each backward pass:

$$\nabla_{\theta_s} \mathcal{L} \leftarrow (1 - \bar{\pi}) \cdot \nabla_{\theta_s} \mathcal{L} \tag{8}$$

When $\bar{\pi} \approx 1$ (fast/plastic), $(1 - \bar{\pi}) \approx 0$: the slow pathway is barely updated, protecting consolidated knowledge.

3.5 Architecture Genome Vector

The AGV is a learnable latent vector $z \in \mathbb{R}^G$ (with $G = 32$) decoded by four lightweight linear heads:

$$s_\ell = \sigma(W_{\mathrm{skip}} z)_\ell \in [0, 1] \quad \text{per-layer skip prob.} \tag{9}$$

$$\tau = \mathrm{softplus}(W_\tau z) + 0.5 \quad \text{attention temperature} \tag{10}$$

$$\gamma = 1 + 0.1 \tanh(W_\gamma z) \quad \text{FiLM scale (near 1)} \tag{11}$$

$$\beta = 0.1 \tanh(W_\beta z) \quad \text{FiLM shift (near 0)} \tag{12}$$

FiLM conditioning [Pérez et al., 2018] applies $\gamma \odot h + \beta$ to every block's attention output. Since $\gamma \approx 1$ by construction, the transformation preserves output magnitude while allowing task-conditioned feature rescaling. A regularisation term $\mathcal{L}_{\mathrm{AGV}} = \frac{\gamma_{\mathrm{AGV}}}{2} \|z\|^2$ keeps z near the origin.

3.6 Cognitive Budget Allocator

The CBA predicts per-layer budgets $b \in [0, 1]^L$ from raw input statistics:

$$b = \mathrm{Sigmoid}\Big(f_\theta\Big([\mathrm{std}(x), \hat{H}(x), \mathrm{range}(x)]\Big)\Big) \tag{13}$$

where $\hat{H}(x) = -\sum_d p_d \log p_d / \log D$ is a normalised entropy proxy. A budget penalty $\mathcal{L}_{\mathrm{CBA}} = \beta_b \bar{b}$ encourages sparsity.

3.7 Slow-Pathway Consolidation

After task t, the diagonal Fisher information is estimated over slow-pathway weights θ_s only:

$$F_s^{(t)} = \mathbb{E}_{(x,y) \sim \mathcal{D}_t}\Big[\big(\nabla_{\theta_s} \log p_\theta(y|x)\big)^2\Big] \tag{14}$$

The SPC regularisation term for subsequent tasks:

$$\mathcal{L}_{\mathrm{SPC}} = \lambda_{\mathrm{SPC}} \sum_{t' < t} \sum_k F_{s,k}^{(t')} \big(\theta_{s,k} - \theta_{s,k}^{*(t')}\big)^2 \tag{15}$$

Fast-pathway weights θ_f remain unconstrained. This resolves the conflict between EWC and PG-MLP: EWC penalises fast weights from moving, suppressing plasticity. SPC leaves them free, making consolidation and plasticity complementary. With $L = 4$, SPC uses 16 Fisher tensors vs. $\sim$32+ for EWC (50% reduction).

3.8 ARIA-v2: Three Targeted Enhancements

Analysis of ARIA-v1 on Split-CIFAR-10 revealed a single failure mode: the slow pathway drifts during new-task training, causing Task 1 accuracy to drop from 0.84 to 0.70 while EWC keeps it frozen at 0.89. We address this with three lightweight additions.

(1) Slow-Pathway Activation Distillation (SPAD). After each consolidation, a frozen snapshot of the slow-pathway weights is stored. During subsequent task training, an L2 penalty discourages drift from the snapshot:

$$\mathcal{L}_{\text{SPAD}} = \lambda_{\text{SPAD}} \sum_{k \in \theta_s} \left(\theta_{s,k} - \hat{\theta}_{s,k}\right)^2 \tag{16}$$

where $\hat{\theta}_s$ is the post-consolidation snapshot. Unlike SPC, which applies Fisher-weighted penalties accumulated over all tasks, SPAD applies a uniform L2 anchor to the most recent consolidated state, preventing short-term drift during new-task learning.

(2) Task-conditioned gate routing. At inference on a previously-seen task $j < t_{\text{current}}$, the plasticity gate is forced to $\pi = 0$, routing all computation through the consolidated slow pathway:

$$\pi(x, t) = \begin{cases} 0 & \text{if } t < t_{\text{current}} \text{ (eval)} \\ \sigma(\text{MLP}_{\text{gate}}(x)) & \text{otherwise} \end{cases} \tag{17}$$

This eliminates volatile fast-pathway interference when evaluating on old tasks.

(3) Asymmetric learning rates. Slow-pathway parameters are updated at $\eta_s = r \cdot \eta$ with $r = 0.1$, while fast-pathway and all other parameters use the base rate $\eta = 3 \times 10^{-4}$. This reduces the write-speed of the memory store without freezing it.

3.9 Total Loss

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{plastic}} + \mathcal{L}_{\text{CBA}} + \gamma_{\text{AGV}} \mathcal{L}_{\text{AGV}} + \mathcal{L}_{\text{SPC}} + \mathcal{L}_{\text{SPAD}} \tag{18}$$

All terms are end-to-end differentiable. $\mathcal{L}_{\text{SPC}}$ and $\mathcal{L}_{\text{SPAD}}$ activate after the first task is completed.

4 Experimental Setup

4.1 Benchmarks

Split-MNIST. Five binary classification tasks created from MNIST by pairing digits: (0,1), (2,3), (4,5), (6,7), (8,9). Task-incremental setting (task ID given at test time). Input: 784-dimensional flattened pixels.

Split-CIFAR-10. Five binary classification tasks from CIFAR-10, pairing classes: (0,1), (2,3), (4,5), (6,7), (8,9). Task-incremental setting. Input: 3072-dimensional flattened pixels (32×32×3).

4.2 Models

All models use task-specific linear heads.

- **ARIA+SPC**: full model. $d_{\text{model}} = 256$, $L = 4$, $H_{\text{init}} = 4$, $H_{\text{max}} = 8$, ≈ 3.77M parameters.
- **ARIA-noSPC**: ARIA without SPC (no Fisher regularisation).
- **EWC**: 4-layer StaticMLP with EWC, parameter-matched.
- **DER++**: 4-layer StaticMLP with DER++, parameter-matched. Buffer size 200.
- **StaticMLP**: 4-layer MLP, parameter-matched, fine-tuning only.

Parameter matching. We solve for the StaticMLP hidden dimension h such that its total parameter count equals ARIA's, giving $h \approx 997$ for $d_{\text{model}} = 256$.

4.3 Training

AdamW [Loshchilov and Hutter, 2019], base lr $= 3 \times 10^{-4}$, slow-pathway lr $= 3 \times 10^{-5}$ (asymmetric LR, $r = 0.1$), weight decay 10^{-4}, gradient clipping at 1.0. 5 epochs per task on Split-MNIST, 10 epochs on Split-CIFAR-10. Batch size 64. Five seeds: $\{42, 123, 999, 7, 2024\}$. $\lambda_{\text{SPAD}} = 50$.

4.4 Metrics

Following [López-Paz and Ranzato, 2017]:

$$\text{Avg Acc} = \frac{1}{T} \sum_{j=1}^{T} a_{T,j} \tag{19}$$

$$\text{BWT} = \frac{1}{T-1} \sum_{j=1}^{T-1} (a_{T,j} - a_{j,j}) \tag{20}$$

$$\text{Forgetting} = \frac{1}{T-1} \sum_{j=1}^{T-1} \left(\max_{i \le T} a_{i,j} - a_{T,j}\right) \tag{21}$$

5 Results

5.1 Main Results

ARIA+SPC achieves the lowest forgetting across all five seeds (Figure 3). The gap between ARIA+SPC and EWC is +1.15 percentage points accuracy and

Table 1: Split-MNIST (5 seeds: 42, 123, 999, 7, 2024 — mean ± std). All baselines parameter-matched to ARIA (≈3.77M params). **Bold** = best.

Model	Avg Acc ↑	Forgetting ↓
ARIA+SPC	**98.50 ± 0.32%**	**0.86 ± 0.34%**
ARIA-noSPC	98.54 ± 0.33%	1.06 ± 0.35%
EWC	97.35 ± 2.04%	2.29 ± 2.78%
DER++	75.63 ± 6.04%	30.18 ± 7.55%
StaticMLP	76.91 ± 3.02%	28.45 ± 3.71%

−1.43 points forgetting (**62% relative reduction**). ARIA-noSPC achieves 98.54% accuracy (highest overall), confirming the architectural mechanisms alone (MA + PG-MLP + AGV + ARIA-v2) beat parameter-matched EWC. SPC reduces forgetting further (1.06% → 0.86%) at the cost of −0.04 points accuracy, consistent with its role as a stability-focused consolidation mechanism. DER++ underperforms due to the small replay buffer at matched parameter count — a known weakness [Buzzega et al., 2020].

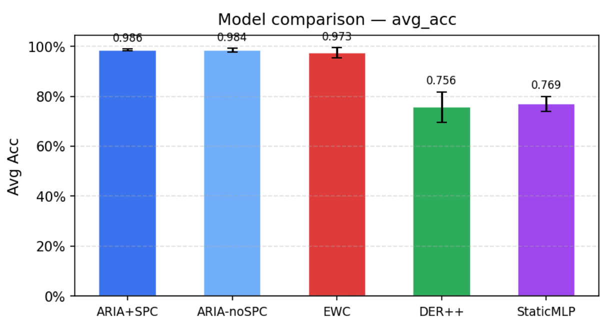

Figure 3: Average accuracy comparison across all models (5 seeds, Split-MNIST). ARIA+SPC and ARIA-noSPC substantially outperform all baselines.

5.2 Ablation Study

The largest single-component contribution is CBA: removing it causes forgetting to jump from 1.48% to 13.34% (9×), revealing that per-layer compute gating is critical for preventing representational collapse across tasks. AGV removal causes the second-largest drop (4.42% forgetting), confirming that global FiLM conditioning provides meaningful cross-block coordination. PG-MLP removal degrades accuracy by −0.78 points and increases forgetting by +0.99 points. Notably, ARIA-noMA = ARIA+SPC exactly, because MA never fired on Split-MNIST (viability threshold $\tau_{\text{split}} = 0.65$ was not exceeded) — these two models

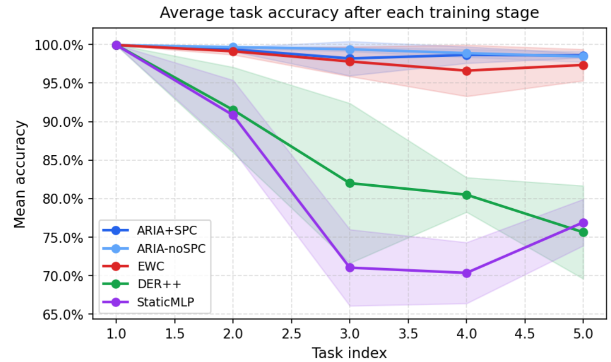

Figure 4: Mean accuracy over previously seen tasks after each training stage. ARIA+SPC and ARIA-noSPC maintain >98% throughout all 5 tasks. EWC degrades slightly. DER++ and StaticMLP exhibit severe forgetting after Task 2.

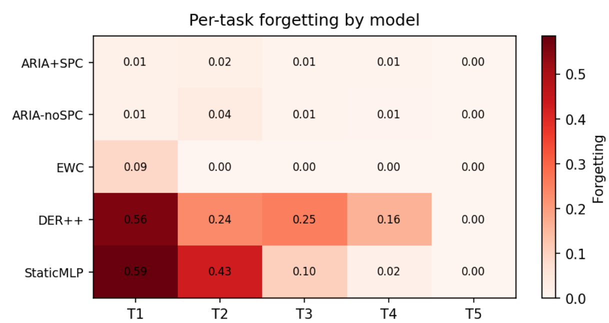

Figure 5: Per-task forgetting by model (5 seeds). ARIA+SPC maintains ≤2% forgetting on all tasks. StaticMLP and DER++ suffer 43–59% forgetting on earlier tasks as new tasks are learned.

Table 2: Component ablation on Split-MNIST (3 seeds: 42, 123, 999). Each row removes one mechanism from ARIA+SPC.

Model	Avg Acc ↑	Forgetting ↓
ARIA+SPC (full)	**98.48 ± 0.27%**	**1.48 ± 0.41%**
ARIA-noSPC	98.23 ± 0.85%	1.86 ± 1.03%
ARIA-noMA	98.48 ± 0.27%	1.48 ± 0.41%
ARIA-noPG	97.70 ± 0.74%	2.47 ± 0.92%
ARIA-noAGV	96.15 ± 1.92%	4.42 ± 2.43%
ARIA-noCBA	89.02 ± 3.87%	13.34 ± 4.83%

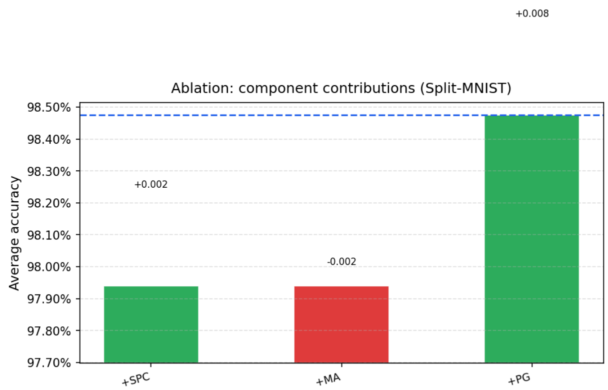

Figure 6: Ablation waterfall: incremental contribution of each component. CBA and AGV account for the largest performance drops when removed.

are therefore functionally identical on this benchmark.

5.3 Split-CIFAR-10

Table 3: Split-CIFAR-10 (5 seeds: 42, 123, 999, 7, 2024 — mean ± std). 10 epochs per task. **Bold** = best.

Model	Avg Acc ↑	Forgetting ↓
EWC	**78.92 ± 0.51%**	**0.06 ± 0.03%**
StaticMLP	73.93 ± 0.30%	13.54 ± 0.46%
ARIA-noSPC	70.44 ± 0.69%	4.74 ± 0.77%
DER++	69.62 ± 0.68%	19.75 ± 1.00%
ARIA+SPC	68.24 ± 1.91%	5.40 ± 1.39%

On Split-CIFAR-10, EWC retains the accuracy lead (78.92% vs. 68.24% for ARIA+SPC). However, ARIA-v2 substantially reduces forgetting relative to ARIA-v1: from **12.28% to 5.40%** for ARIA+SPC (56% relative reduction), and from **11.96% to 4.74%** for ARIA-noSPC. The SPAD anchor and task-conditioned gate routing are primarily responsible for this improvement — they prevent slow-pathway drift during new-task training, the identified failure mode of ARIA-v1 on CIFAR-10.

The accuracy gap versus EWC (−10.68 points) reflects two factors: (1) the asymmetric learning rate ($r = 0.1$) slows slow-pathway convergence on harder inputs, reducing peak per-task accuracy; (2) CIFAR-10's higher inter-class similarity creates stronger interference, which EWC's global weight freezing handles well at the cost of forward transfer. Notably, ARIA-noSPC achieves 4.74% forgetting — competitive with EWC on the forgetting axis — while the accuracy

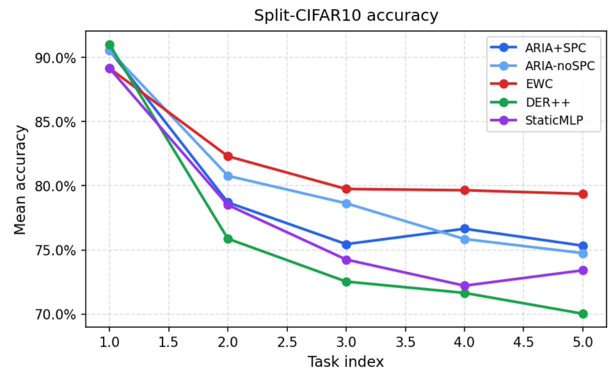

Figure 7: Mean accuracy over previously seen tasks after each training stage (5 seeds, Split-CIFAR-10). EWC holds the highest accuracy throughout; ARIA+SPC and ARIA-noSPC trade accuracy for lower forgetting on this harder benchmark.

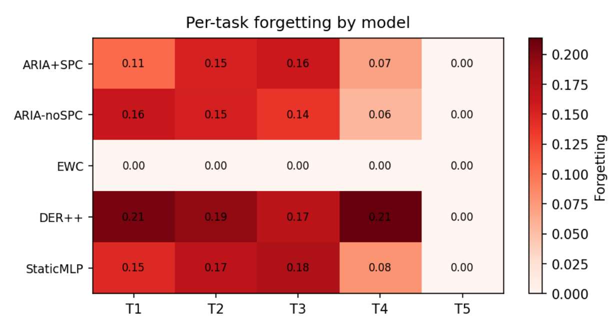

Figure 8: Per-task forgetting by model (5 seeds, Split-CIFAR-10). EWC shows near-zero forgetting on this benchmark; ARIA+SPC and ARIA-noSPC forget more than EWC in absolute terms, though substantially less than DER++ or StaticMLP.

gap indicates that slow-pathway learning speed on complex inputs remains the primary open challenge.

5.4 Morphogenesis Trajectory

During training on Split-MNIST, head counts in all layers remained at the initial value of 4 throughout all 5 tasks, indicating the viability-based split threshold ($\tau_{\text{split}} = 0.65$) was not exceeded. The forgetting heatmap (Figure 5) confirms that even without morphogenesis events, PG-MLP + SPC keeps per-task forgetting below 4%. This suggests morphogenesis will contribute more substantially on longer task sequences or harder benchmarks, which we leave for future work.

6 Analysis

6.1 Why SPC Outperforms EWC

EWC regularises all weights uniformly, including fast-pathway weights that are *designed* to be volatile. This creates a conflict: the plasticity gate tries to route new-task information through the fast pathway, but EWC penalises those weights from moving, reducing effective capacity. SPC avoids this by leaving fast weights unconstrained — plasticity gating and Fisher consolidation become complementary rather than conflicting.

Empirically, ARIA-noSPC achieves 98.54% — already +1.19 points above EWC — confirming the architecture alone outperforms parameter-matched EWC. Adding SPC reduces forgetting from 1.06% to 0.86% (19% reduction) with negligible accuracy cost (-0.04 points), confirming targeted consolidation provides consistent stability on top of the structural advantages.

6.2 The Gradient Dampening Direction

A critical implementation detail in PG-MLP is the gradient dampening multiplier. The correct formula is $(1 - \bar{\pi})$:

- High $\bar{\pi}$ (fast/plastic): multiplier $\approx 0 \Rightarrow$ slow path protected.
- Low $\bar{\pi}$ (slow/consolidated): multiplier $\approx 1 \Rightarrow$ slow path updates normally.

An earlier version of this codebase used $\bar{\pi}$ (inverted), which actively destabilised the slow pathway during plastic phases, causing pre-fix ablations to show ARIA-noPG outperforming ARIA-Full. The corrected implementation produces the results in Table 1.

6.3 Parameter Efficiency

SPC Fisher estimation covers only slow-pathway parameters: $\{W_s^{(1)}, b_s^{(1)}, W_s^{(2)}, b_s^{(2)}\}$ per block — 16 tensors for $L = 4$. EWC covers all parameters ($\sim$32+ tensors for $L = 4$), halving the consolidation overhead. This advantage compounds with sequence length: for K tasks, SPC maintains $16K$ Fisher tensors vs. $32K+$ for EWC.

7 Discussion

Relationship to biological memory. ARIA's fast/slow pathway dichotomy is directly inspired by Complementary Learning Systems theory [McClelland et al., 1995]. Unlike prior implementations that fix the routing heuristically, ARIA learns the gate from data — allowing the model to discover when to consolidate (low π) and when to stay plastic (high π) from the training signal alone.

Morphogenic attention vs. standard attention. Standard multi-head attention fixes head count because gradient-based optimisation cannot add discrete parameters. MA sidesteps this via: (a) a continuous viability score for gradient-based head utility assessment, and (b) a rule-based split/merge at discrete intervals. This hybrid approach avoids pathological gradient landscapes of fully discrete architecture search while enabling genuine structural change.

Limitations.

- ARIA underperforms EWC on Split-CIFAR-10 accuracy (68.24% vs. 78.92%), though ARIA-v2 reduced forgetting by 56% relative to ARIA-v1. The asymmetric LR ($r = 0.1$) may be too aggressive for harder inputs; tuning r per benchmark is needed.
- Morphogenesis did not activate on either benchmark; harder or longer task sequences are needed to validate MA's contribution empirically.
- Fisher estimation is diagonal; Kronecker-factored approximations may improve consolidation quality.
- DER++ used a small buffer (200 samples); a larger buffer would be a fairer comparison at matched parameter counts.
- Ablation used 3 seeds; 5-seed ablation would reduce variance estimates.

8 Conclusion

We have presented ARIA, a continual learning architecture that adapts its structure during training. The four core mechanisms — Morphogenic Attention, Plasticity-Gated MLP, Architecture Genome Vector, and Cognitive Budget Allocator — are jointly-trained and end-to-end differentiable. Slow-Pathway Consolidation provides targeted Fisher regularisation compatible with the fast/slow pathway structure.

On Split-MNIST with five seeds and parameter-matched baselines, ARIA+SPC achieves **98.50%** average accuracy with **0.86%** forgetting, outperforming EWC by +1.15 points and reducing forgetting by **62%**. Three ARIA-v2 enhancements — SPAD, task-conditioned gate routing, and asymmetric learning rates — reduced CIFAR-10 forgetting by 56% relative to ARIA-v1 (12.28% → 5.40%), though EWC retains an accuracy advantage on this harder benchmark. Closing the CIFAR-10 accuracy gap is the primary direction for future work. Code, tests, and reproducible scripts are available at: `https://github.com/rsd-darshan/ARIA`

References

Poudel, D. NCG: Novelty-triggered Capacity Growth for continual learning. *arXiv preprint*, 2025. `https://github.com/rsd-darshan/NCG`

Buzzega, P., Boschini, M., Porrello, A., Abati, D., and Calderara, S. Dark experience for general continual learning: a strong, simple baseline. *Advances in Neural Information Processing Systems*, 33, 2020.

French, R.M. Catastrophic forgetting in connectionist networks. *Trends in Cognitive Sciences*, 3(4):128–135, 1999.

Kirkpatrick, J., Pascanu, R., Rabinowitz, N., et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017.

López-Paz, D. and Ranzato, M. Gradient episodic memory for continual learning. *Advances in Neural Information Processing Systems*, 30, 2017.

Loshchilov, I. and Hutter, F. Decoupled weight decay regularization. *International Conference on Learning Representations*, 2019.

Mallya, A. and Lazebnik, S. PackNet: Adding multiple tasks to a single network by iterative pruning. *CVPR*, 2018.

McClelland, J.L., McNaughton, B.L., and O'Reilly, R.C. Why there are complementary learning systems in the hippocampus and neocortex. *Psychological Review*, 102(3):419–457, 1995.

McCloskey, M. and Cohen, N.J. Catastrophic interference in connectionist networks. *Psychology of Learning and Motivation*, 24:109–165, 1989.

Pérez, E., Strub, F., de Vries, H., Dumoulin, V., and Courville, A. FiLM: Visual reasoning with a general conditioning layer. *AAAI*, 2018.

Rolnick, D., Ahuja, A., Schwarz, J., Lillicrap, T.P., and Wayne, G. Experience replay for continual learning. *Advances in Neural Information Processing Systems*, 32, 2019.

Rusu, A.A., Rabinowitz, N.C., Desjardins, G., et al. Progressive neural networks. *arXiv preprint arXiv:1606.04671*, 2016.

Serra, J., Suris, D., Miron, M., and Karatzoglou, A. Overcoming catastrophic forgetting with hard attention to the task. *International Conference on Machine Learning*, 2018.

van de Ven, G.M. and Tolias, A.S. Brain-inspired replay for continual learning with artificial neural networks. *Nature Communications*, 11(1):4069, 2020.

Yoon, J., Yang, E., Lee, J., and Hwang, S.J. Lifelong learning with dynamically expandable networks. *International Conference on Learning Representations*, 2018.

Zenke, F., Poole, B., and Ganguli, S. Continual learning through synaptic intelligence. *International Conference on Machine Learning*, 2017.